



Mark Scheme (Results)

June 2024

Pearson BTEC Nationals in IT

Unit 1: Information Technology Systems (31760H)

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Unit 1: Information Technology Systems

General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- All marks on the mark scheme should be used appropriately.
- All marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if a candidate's response is not worthy of credit according to the mark scheme.
- Where some judgment is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt about applying the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed-out work should be marked UNLESS the candidate has replaced it with an alternative response.
- Phonetic spelling should be accepted.

Specific marking guidance

The marking grids have been designed to assess learner work holistically. Rows in the grids identify the assessment focus/outcome being targeted. When using a marking grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer, in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band, depending on how they have evidenced each of the descriptor bullet points.

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Question Number	Answer	Mark
1ai	Award one mark for each response from: Barcode reader/scanner (1) QR reader (1) Keyboard (1) Mouse (1) Card reader (1) Touch screen (1) DO NOT ACCEPT: Tablet or scanner on its own Accept any other appropriate response.	2

Question Number	Answer	Mark
1aii	Award one mark for the identification and one additional mark for the appropriate expansion to a maximum of four marks: • Improve efficiency (1) because stock data is in real time (1) • More accurate (1) because there is less risk of human error (1) • Saves time (1) because it will trigger orders when the minimum levels are reached (1) • Reduced costs (1) because it reduces overstocking of books / the need to employ someone to manage stock (1) Accept any other appropriate response.	4

Question Number	Answer	Mark
1bi	Award one mark for the identification and one additional mark for the appropriate expansion to a maximum of four marks: • More convenient (1) because the software has already been developed (1) • Reliability (1) because the software is already in widespread use/tried and tested (1) • Reviews can be obtained prior to purchasing (1) because it will give an insight into the functionality of the software (1) • Regular updates are available from the manufacturer (1) who will improve functionality/fix existing issues such as bugs/patches in the software (1) • Cheaper than bespoke software (1) because bespoke software is created to order/suited to individual needs (1) • Troubleshooting/support/community (1) which will solve any problems the software may have (1) • Some software companies offer a free trial option (1) so the owner can try out the product before purchasing a suite of applications (1) Accept any other appropriate response.	4

Question Number	Answer	Mark
1bii	Award one mark for the identification and one additional mark for the appropriate expansion to a maximum of four marks: Additional costs (1) which can be due to annual license / upgrade costs / if customer requires 24/7 support (1) Limited customisation (1) because there is no access to the source code/ the developer of the software has control over its development and features (1) Compatibility issues (1) because not all hardware and software will be able to support off the shelf applications software / will need to purchase new devices (1) Features/functions may be limited (1) which will result in the off-the-shelf software not meeting all the bookstore's requirements / could reduce competitive advantage (1) Accept any other appropriate response.	4

Question Number	Indicative content
1c	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners discuss the factors that will affect the choice and use of other user interfaces. • Ease of use. How simple and intuitive the user interface is to navigate. • User experience. Graphical user interfaces are easier to use/learn for the more inexperienced user. • Accessibility. Is the interface designed for people with additional needs? • Performance. How quickly the user interface responds to user input. Some user interfaces have slow response times, e.g. speech, because the software will need to process the user's voice and could take a long time to interpret. • User requirements. Does the user interface meet requirements and functionality for what is needed to do? • Storage space. Is the storage space sufficient to handle the size of the interface? Some need a significant amount of storage to run. <u>Graphical:</u> • Much easier to use for beginners • Can exchange information between software easier • Use lots of memory and processing power, which can result in the interface being slower loading • Attractive and pleasing to the eye for the user with the use of colours • Easy to use <u>Command line:</u> • Allows interaction with the computer by typing in commands • Commands must be typed correctly otherwise they might not work • Quicker to use for experienced technical users who know commands, whereas the inexperienced user will find it difficult to use because of the number of commands that are required to learn. • Memory and processing power is not as important, therefore will run on lower specification machines. • No need to install Windows software on the computer <u>Menu based:</u> • Easy to use because the user does not have to remember commands • User friendly because the user can navigate around the interface until they find what they are looking for • Can be annoying if there are many different levels of menus to move around

Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	3–4	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	5–6	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Answer	Mark
2a	Award one mark for identification and one additional mark for appropriate expansion. • Secure (1) because to gain access the user has to have physical access to a network port (1) • Reliable/stable (1) because signal strength does not deteriorate (1) • Fast data transfer speeds (1) because they are less prone to interference (1) Do not accept: 'safer' for 'secure' Accept any other appropriate response.	2

Question Number	Answer	Mark
2b	Award one mark for the identification and one additional mark for the appropriate expansion to a maximum of four marks: • (The operating system) multi-tasking (1) which will support the users running multiple applications at the same time (1) • A multi-user operating system supports concurrent access (1) so multiple users can access the system at the same time (1) • Resource sharing (1) so the staff can share peripherals such as printers, hard drives (1) • Background processes (1) commands are executed in the background not in real-time (1) Accept any other appropriate response.	4

Question Number	Answer	Mark
2c	Award one mark for the identification and one additional mark for the appropriate expansion to a maximum of four marks: • Communication between the operating system and devices (1) through a set of interfaces and protocols (1) • Translates/interprets instructions (for the device to handle) (1) to ensure the devices work correctly/efficiently (1) • Device drivers provide commands that control peripheral devices (1) so the operating system does not have to be programmed with commands for each device (1) Allow: mix and match Accept any other appropriate response.	4

Question Number	Answer	Mark
2d	Award one mark for each up to a maximum of four marks. • Firewall forms a barrier between the device and the network (1) • Using either hardware and/or software (1) • Connected using designated ports (1) • Filters / checks incoming/outgoing traffic (1) • Against a set of rules / database rules (1) • Allows only authorised traffic to access the network (1) Allow: Reverse of MP6 (blocks unauthorised traffic) MP5 examples of database rules, e.g. IP addresses, port numbers Accept any other appropriate response.	4

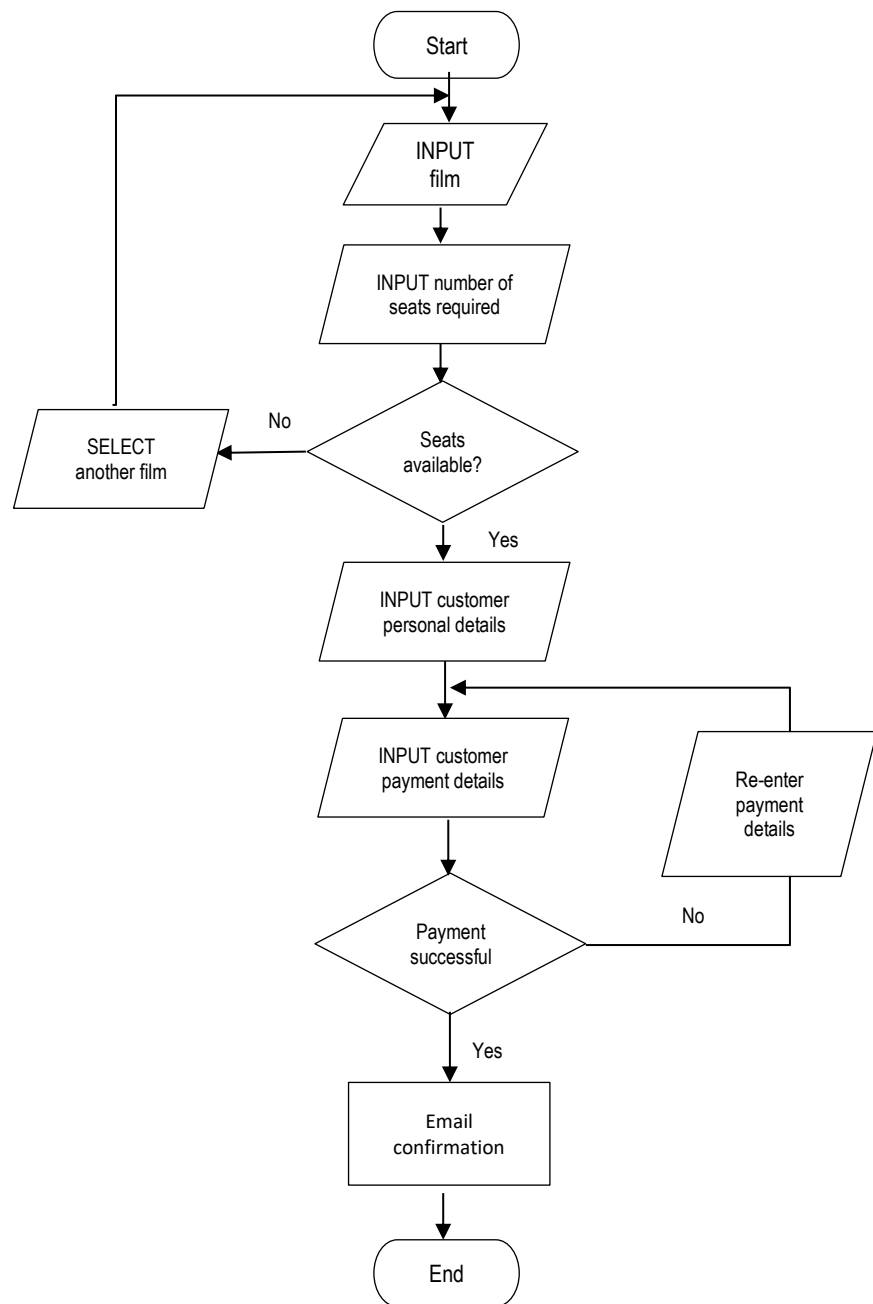
Question Number	Answer	Mark
2e	Award one mark for the identification and one additional mark for the appropriate expansion to a maximum of four marks: Backup: To ensure there is a copy of data (1) in the event of data loss/corruption (1) Can provide an automated solution for creating/managing/restoring data (1) which will reduce the need for manual backups / human error (1) Compression: (Data compression) reduces the size of the file (1) because it takes up less space on the drives / faster data transfer / to enable email attachments / transfer of larger files (1)	4

Question Number	Indicative content
2f	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Discuss methods, other than firewalls, that can be used to keep data secure on the network. Encryption • So only authorised people can read it • Uses an encryption key • Requires a decryption key to read the data • Two main types of encryption: ◦ Asymmetric (public-key encryption) ◦ Symmetric Virus protection • Designed to prevent viruses, worms and Trojan horses from getting into a computer • Removes malicious software that has already infected a computer Physical access controls • Physical access to systems. The employees must take responsibility for making sure they log off the computers when not in use, locked in a secure room. • External storage devices must be used with caution, and the same physical access controls must be applied, e.g. passwords, encryption, etc.

Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1-3	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	4-6	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	7-8	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Indicative content
3a	A flow chart OR diagram showing the process for customers to book film tickets online by completing a booking form to meet the requirements of the scenario. Candidates do not need to use any standard symbols or conventions. Credit should be given for a flow chart that meets the requirements of the given problem. Example response – note this is indicative only. Flow chart should include logic for: • Input the film • Input the number of seats required ○ are there enough seats available? ○ if no output error message ○ loop back to input film • Input personal details • Input payment details ○ is payment successful? ○ if no output error message ○ loop back to payment details • If all conditions are met confirmation email sent See attached flow chart for sample solution. CORRECT SYMBOLS used: No need to specify the word 'Input' or 'output' or indicate a decision. INCORRECT SYMBOLS used: Learners must specify 'Input', 'output' or indicate a decision in the relevant boxes.



Level	Descriptor
0 marks	No rewardable material
1-2 marks	Diagram provides partial coverage of appropriate decisions and shows some logical structure to meet the requirements of the scenario. Diagram includes annotations to show the results of decision making.
3-4 marks	Diagram provides coverage of mostly appropriate decisions and logical structure to meet the majority of the requirements of the scenario, but the structure might not be the most efficient. Diagram includes accurate annotations and technical language to show the results of decision making.
5-6 marks	Diagram provides thorough coverage of appropriate decisions and logical structure to fully meet the requirements of scenario. Diagram includes accurate annotations and technical language to accurately represent an efficient system.

Question Number	Answer	Mark
3b	Award one mark for the identification and one additional mark for the appropriate expansion to a maximum of four marks: Hypertext Transfer Protocol Secure (HTTPS) (1) authenticates the accessed website / to add a layer of security (on the data in transit) (1) Secure Socket Layer (SSL) / Transport Layer Security (TLS) (1) establishes a secure link between a web server and browser (1) Public Key Infrastructure (PKI) / Secure Electronic Transaction (SET) (1) enables an unsecure public network to securely exchange data / manages digital certificates / digital signatures / encryption (1) Allow: Mix and match if it makes sense, but not the same expansion for MP1 and MP2 once only. Do Not Accept: To keep payment details secure/transfer securely (in the question). Accept any other appropriate response.	4

Question Number	Answer	Mark
3c	A description of how verification and validation may be used in a password field such as: Verification The password will be entered twice/asked to be re-entered (1) compared to ensure they match (1) Validation The password will be checked against pre-set rules (1) to ensure entries follow the rules (1) ALLOW: Examples of pre-set rules, e.g. length, mix of characters, upper and lowercase letters, no personal details. Accept any other appropriate response.	4

Question Number	Indicative content
3d	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners will discuss the legal issues and implications for the cinema and its staff when using IT systems. <u>display screen equipment (DSE) regulations</u> • Undertake a DSE workstation assessment including equipment, furniture and any special requirements of a member of staff • Make sure staff take breaks from DSE to reduce risks • Provide an eye test if a staff member asks for one, and provide glasses if a staff member needs them ○ Staff can help they eyes by checking the screen is well positioned/adjusted, lighting conditions are suitable, take regular breaks from screen work • Provide training and information for staff ○ good posture ○ adjusting chairs and other furniture ○ arranging desk space ○ adjusting screens and lighting to avoid reflections and glare ○ breaks and changes of activity ○ risk assessments ○ how to report problems <u>equality legislation</u> • protects people from discrimination in the workplace, e.g. gender, race, disability, etc. • prohibited conduct includes: ○ direct and indirect discrimination ○ discrimination arising from disability ○ adjustments for disabled persons ○ harassment ○ victimisation <u>disability discrimination legislation</u> • Rights to protect individuals from discrimination includes employment, education and dealing with the police. • Against the law for employers to discriminate because of a disability and the Equality Act protects individuals for: ○ Application forms ○ Interview arrangements ○ Aptitude/proficiency tests ○ Job offers ○ Terms of employment, including pay ○ Promotion, transfer and training opportunities ○ Dismissal or redundancy ○ Discipline and grievances • Employers must make reasonable adjustments in the workplace

	- Employers can only make limited enquiries about a potential staff member's health or disability: - to help decide if the employee can carry out a task that is essential to the job role - to help find out if the employee can take part in an interview - to help decide if the interviewers need to make reasonable adjustments - if they want to increase the number of disabled people they employ - if they need to know for purposes of national security checks - Employees cannot be chosen for redundancy because they are disabled - Employees cannot be forced to retire if you become disabled
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Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1-3	- Demonstrates isolated elements of knowledge and understanding - There will be major gaps or omissions - Few of the points made will be relevant to the context in the question - Limited discussion which contains generic points - Little or no consideration of different aspects
Level 2	4-6	- Demonstrates some accurate knowledge and understanding - There will be some gaps or omissions - Some of the points made will be relevant to the context in the question, but the link will not always be clear - Displays a partially developed discussion which considers some different aspects - There will be some consideration of how they interrelate
Level 3	7-8	- Demonstrates mostly accurate and detailed knowledge and understanding - There will be minor gaps or omissions - Most of the points made will be relevant to the context in the question, and there will be clear links - Displays a well-developed and logical discussion which clearly considers a range of different aspects - Clearly shows how they interrelate

Question Number	Answer
4a	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners discuss the features, use and purpose of different networks. LAN • Every computer on the network can communicate with any of the other computers on the network using a single internet connection • There is a high degree of interconnection between computers • It is easy to connect computers • Data transmission is inexpensive • High data transmission rates • Usually connects devices in relatively close proximity • Often connected using wired connections • Access to centralised servers that house applications and data • Allow resource sharing • Network security tools can be used to protect LAN connected devices • Network topologies: star, ring, mesh, bus, tree • Each separate regional office can be set up using a LAN WAN • Collection of LANs that communicate with one another, e.g. a network of networks • Networking over a wider area, e.g. whole city, country, continent, etc. • Multiple LAN networks are used and connected to a WAN • The company can connect all regional office LANs to the WAN • Regional offices can all share data and information across the network with high speed and efficiency • TCP/IP suite of communication protocols that are used to interconnect network devices • Router is a networking device that is used to interconnect LANs to form a WAN. IP routers use IP addresses to determine where to send packets of data. Each device is assigned an IP address.

Mark scheme (award up to 10 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–4	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	5-7	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	8-10	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Indicative content
4b	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners analyse the characteristics and implications of SSDs and HDDs. <u>Price</u> - SSDs are more expensive compared to HDDs. HDDs are an older and more established technology. <u>Capacity</u> - SSDs have a smaller capacity than HDDs - High capacity SSD units are rare and expensive - HDDs have high capacities (250-500GB considered standard) - Civil engineering users will require more capacity, due to their need for space to store designs, drawings, etc. <u>Speed</u> - Systems with SSDs are much faster to boot than with HDDs which take time to speed up due to read/write head movement and seek times. - SSDs load applications faster and have higher performance. - Transfer times are quicker <u>Weight</u> - If using laptops with SSDs, they are generally lighter due to the lack of mechanical parts. <u>Fragmentation</u> - HDDs operate well with large files that are saved in continuous blocks because they spin and it allows the read head to work in one continuous motion. - When HDDs fill up continuous blocks are less available and files are distributed across the disk surface which will slow down file loading times. - SSDs store data on chips, and with no physical read head ensures read speeds are consistent across the drive. <u>Durability</u> - HDDs have moving parts which means they can be damaged if dropped or shaken while operating. - HDDs have read/write heads that pass across the drive at high speed and may come into contact with various micro-debris which will cause loading issues. - SSDs are not prone to errors. <u>Availability</u> - SSDs are becoming more popular, however HDDs are far more common.

	<u>Form factors</u> • HDDs have spinning platters, which makes them bigger than SSDs • SSDs are becoming smaller every year which is resulting in many laptops and tablets are becoming slimmer, adopting SSDs. <u>Noise</u> • SSDs are not mechanical therefore make them quieter and almost silent. • HDD drives make noise when they spin, or when their read arm is operational.
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Mark scheme (award up to 12 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1-4	• Demonstrates isolated knowledge and understanding, there may be major gaps or omissions • Breaks the situation down into component parts and a few of the points made will be relevant to the context in the question • Limited analysis which contains generic assertions rather than interrelationships or linkages
Level 2	5-8	• Demonstrates some accurate knowledge and understanding, with few minor omissions/any gaps or omissions are minor • Breaks the situation down into component parts and some of the points made will be relevant to the context in the question • Displays a partially developed analysis which considers some interrelationships or linkages but not always sustained
Level 3	9-12	• Demonstrates mostly accurate and thorough/detailed knowledge and understanding • Breaks the situation down into component parts and most of the points made will be relevant to the context in the question • Displays a well-developed and logical analysis which clearly considers interrelationships or linkages in a sustained manner



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Welsh Assembly Government

